A Human–AI Readout Direction Changes Perceived Writing Style but Fails Preservation Tests

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Abstract

Does a direction that separates human and machine text also control the style of new generations? We test this question in paired base and instruction Qwen2.5-1.5B checkpoints using source-matched continuations, signed residual additions, projection ablation, five equal-norm random controls, and nuisance residualization. We evaluate 1,472 generated task outputs with a frozen encoder detector and a blinded external judge, alongside coherence, factual coverage, and unsupported claims. At the validation-selected layer, negative steering reduces judge-rated AI-likeness by 20.04 points on a 100-point scale (95% paired bootstrap interval: $[-22.85, -17.27]$). However, coherence falls by 0.591 points on a five-point scale, and neither prespecified preservation criterion passes. The encoder does not corroborate the primary effect; positive steering increases judge-rated AI-likeness while decreasing its detector score. Readout is strong in both base and instruction backbones (held-out AUROC 0.9707 and 0.9704), but the instruction-backbone direction fails to transfer to base-generator continuations (0.486). Removing five nuisance proxies retains a smaller judged style effect with continued quality loss. These results establish partial control of perceived writing style under this protocol, while leaving a distinct, detector-consistent, content-preserving direction unestablished. They show why readout, controllability, measurement validity, and preservation require separate tests.

1 Introduction

A model can recognize patterns in machine-written text without using the same patterns to produce its own writing. Distinguishing these two possibilities matters for interpreting representation-based detectors: an accurate linear readout does not by itself identify a controllable writing mechanism. Even when an intervention lowers an AI-style score, it may do so by damaging coherence, removing facts, or exploiting weaknesses in the evaluator.

Can reading out AI style predict control of new writing? Linear probes and mean-difference detectors extract substantial human–machine information from frozen language-model activations [Quaremba et al., 2026, Vishnyakov and Gaintseva, 2026]. Prior sparse-autoencoder work also steers individual stylistic features, with factual side effects at strong doses [Kuznetsov et al., 2025]. The unresolved question is therefore more specific than whether style can be steered: can a human–AI readout direction change independent judgments of new text while preserving coherence and content, and can this effect be distinguished from nuisance style and persona factors?

We test this conjunction in paired pretrained and instruction-tuned QWEN2.5-1.5B checkpoints. We fit directions to human and GPT-4o continuations from matched source documents, select a layer on validation readout accuracy, and intervene during generation on 64 unseen factual records. Figure 1 summarizes the protocol. Signed doses, projection ablation, equal-norm random directions, nuisance controls, and a plain human-style prompt separate several explanations for a score change. A frozen RoBERTa detector and a blinded Gemini judge evaluate the outputs; explicit noninferiority margins test preservation.

Negative steering lowers judged AI-likeness by 20.04/100 points, but reduces coherence by 0.591/5 points and fails both preservation gates. The encoder change is $+0.061$, with a confidence interval spanning zero. Strong readout in both backbones also fails to generalize to Llama-3 base continuations. We make three contributions:

Research manuscript; formatted with the NeurIPS 2025 style.

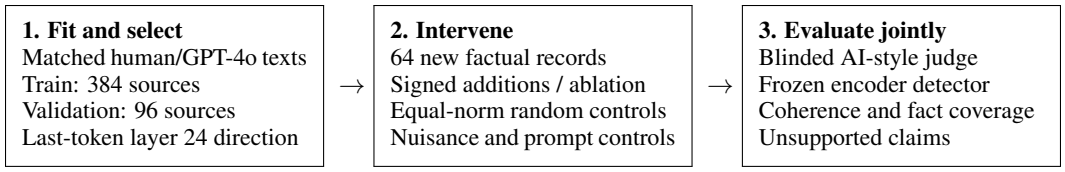


Figure 1: Readout-to-generation test. Layer selection uses held-out readout accuracy, without generation scores. We pair generation outcomes by factual record and test style alongside preservation. A style-score change alone does not satisfy the hypothesis.

- We test whether a source-matched human–AI residual readout controls new generations using signed, random, ablation, nuisance, and prompt interventions.
- We quantify detector disagreement and quality costs with paired inference, calibration, and prespecified preservation margins.
- We compare domain and generator readout transfer across paired base and instruction backbones, documenting a base-generator transfer failure and no observed amplification in AUROC.

Section 2 positions the study; section 3 defines the protocol; section 4 reports the results; and section 5 examines their limits.

2 Related Work

Representation-based detection. Quaremba et al. use last-token residual features and linear probes to detect machine-generated text across evaluation settings [Quaremba et al., 2026]. SV-DETECT estimates layerwise directions and combines their features for classification [Vishnyakov and Gaintseva, 2026]. These methods establish useful readout structure, but their detection results do not establish generation control. We use a single-layer direction in native residual coordinates, enabling a direct intervention rather than treating a multi-layer detector as one physical vector.

Causal steering and persona. Contrastive Activation Addition averages paired residual differences and applies signed interventions to generation [Rimsky et al., 2024]. Kuznetsov et al. already steer sparse-autoencoder decoder features associated with complexity, verbosity, formality, and repetition [Kuznetsov et al., 2025]. Their factual side effects motivate our quality tests; we do not claim the first causal AI-style intervention. The Assistant Axis characterizes and intervenes on assistant persona across many roles [Lu et al., 2026]. We include an assistant-versus-narrator output-text proxy in the tested model, but this reduced control does not reproduce that multi-role axis. Our contribution is the combined intervention, measurement, and preservation test, including its negative findings.

Matched corpora and distribution shift. HAP-E compares human and model continuations from shared context, revealing grammatical and rhetorical differences across generators and domains [Reinhart et al., 2025]. HC3 matches human and ChatGPT answers by question [Guo et al., 2023]. Such matching controls context or topic, not every factual proposition. RAID and MAGE document difficulties under changes in generators, domains, decoding, and attacks [Dugan et al., 2024, Li et al., 2024]. We accordingly separate instruction- and base-generator transfer. Fast-DetectGPT provides a likelihood-based alternative to encoder detection [Bao et al., 2024]; we did not run it. Instead, we evaluate a historical encoder and an external style judge, and explicitly test their validity under the study's distribution shift.

3 Methodology

3.1 Question, models, and data

We operationalize "sounds like AI" as a measured property of text, not a claim about authorship. The primary estimand is the prompt-paired change under negative steering in each of two external style scores. The stronger hypothesis additionally requires preserved coherence and factual coverage and evidence against the tested nuisance explanations.

We use QWEN2.5-1.5B base and instruction checkpoints with frozen weights [Qwen Team, n.d.a,n]. These historical 1.5B checkpoints permit paired interventions with accessible internals; they are not contemporary capability leaders. Section A provides revisions and runtime details.

Readout corpus. From HAP-E, we join human chunk 2 and generated continuations by source ID; human chunk 1 is context, not a comparison response. Academic, blog, and news sources provide 384 training, 96 validation, and 128 test documents. A further 128 fiction, spoken, and TV/movie sources test domain transfer. Fitting uses GPT-4o versus human text only. Each test source also includes Llama-3-8B base and instruction continuations, sharing the same human reference. Another 128 HC3 question pairs test task transfer without fitting. Together these sets contain 2,240 texts. We retain the first 128 whitespace-delimited words and at most 192 model tokens; this approximates rather than guarantees length matching.

Source IDs and normalized hashes are disjoint across folds. A training-vocabulary character-4-gram TF–IDF audit finds maximum test-to-training cosine 0.580 and no values above 0.90. These checks cannot rule out semantic overlap or pretraining contamination.

Generation tasks. We construct 64 fictitious factual records: eight variants in each of eight community-event topics. Prompts request a 70–100-word paragraph preserving the location, event, day/time, coordinator, item count, free attendance, and registration deadline, without extra claims. The records are procedural test inputs; all model outputs come from actual inference. The shared templates limit the diversity of the 64 inferential units.

3.2 Direction fitting and generation intervention

We pass raw response text without chat framing through each checkpoint and extract the last real token after blocks 8, 16, and 24 (one-based), before final normalization. Left padding places that token at the final position. For layer ℓ, the unit AI-minus-human direction is

$$\boldsymbol{v}_\ell = \frac{\bar{\boldsymbol{h}}_{\ell,\mathrm{AI}} - \bar{\boldsymbol{h}}_{\ell,\mathrm{human}}}{\|\bar{\boldsymbol{h}}_{\ell,\mathrm{AI}} - \bar{\boldsymbol{h}}_{\ell,\mathrm{human}}\|_2}. \tag{1}$$

We also fit an L2 logistic normal ($C = 1$, maximum 2,000 iterations) in native coordinates, without PCA or feature scaling. Layer 24 maximizes validation mean-direction AUROC for both checkpoints: 0.9811 instruction and 0.9825 base. Generation scores play no role in layer selection.

For a unit intervention vector $\boldsymbol{u}$, we apply

$$\boldsymbol{h}' = \boldsymbol{h} + \alpha\sigma\boldsymbol{u}, \qquad \sigma = \mathrm{SD}_{\mathrm{train}}(\boldsymbol{h}^\top \boldsymbol{v}), \tag{2}$$

where $\boldsymbol{v}$ is the original selected mean direction. The instruction/base scales are 18.811/19.704. All alternative addition directions use this same absolute scale, not their own projection standard deviation. At $|\alpha| = 2$, additions have roughly 23–24% of the mean training last-token residual norm. Projection ablation instead uses the uncentered operation $\boldsymbol{h}' = \boldsymbol{h} - (\boldsymbol{h}^\top \boldsymbol{v})\boldsymbol{v}$; it is not negative addition.

The hook changes the last prompt token predicting the first output and each subsequent cached generation token. Earlier prompt states remain unchanged. Arithmetic, earlier-position invariance, zero-hook equivalence, and repeated-sampling smoke tests pass for both checkpoints. These checks validate implementation, not the causal suitability of the layer.

Conditions and sampling. The instruction model has 17 conditions: zero; mean $\alpha \in \{-2, -1, +1, +2\}$; logistic -2; projection ablation; five random -2 directions; formality -2; assistant proxy -2; residualized mean ± 2; and a plain "write like a human" prompt. All use seed 42; zero and mean ± 2 also use seed 43. The base model runs zero and mean ± 2 at seed 42. This yields $64(17 + 3 + 3) = 1{,}472$ task outputs. We use temperature 0.7, top-p 0.9, top-k 0, and at most 160 new tokens. Batch seeds add the starting batch offset. Instruction generation uses the chat template; base generation uses `Instruction: ... Response:`. This difference prevents attributing generation contrasts solely to instruction tuning.

3.3 Nuisance controls and independent measurements

For each checkpoint, we greedily generate 24 pairs for each of formal/casual, verbose/concise, and helpful-assistant/fictional-narrator prompts on separate pottery-course records: 144 outputs per

checkpoint. Last-token mean differences give three proxies. The supplied facts are fixed, but generated realizations may differ in content. Training activation/length covariance and an academic-versus-other-domain mean difference supply two further proxies. With an orthonormal QR basis $\boldsymbol{Q}$ for these five vectors, we use

$$\widetilde{\boldsymbol{v}} = (\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\boldsymbol{v}, \qquad \boldsymbol{v}_{\text{res}} = \widetilde{\boldsymbol{v}}/\|\widetilde{\boldsymbol{v}}\|_2. \tag{3}$$

This removes a measured linear span, not all semantic influence of those constructs.

We score generated text with the frozen `Hello-SimpleAI/chatgpt-detector-roberta` ChatGPT probability [Hello-SimpleAI, n.d.] and a Gemini judge (`google/gemini-3.8-flash`). Style scoring excludes prompts and intervention labels; quality scoring additionally receives the source facts. The judge rates AI-likeness (0–100), coherence (1–5), source-fact coverage (0–1), and unsupported-claim fraction (0–1). Requests shuffle conditions into batches of eight with opaque integer IDs. The measures share neither fitted activations nor condition labels; we do not assume independent errors.

Because the encoder model card documents training on all HC3, we calibrate both measures on 64 held-out HAP-E human/GPT-4o pairs, not HC3. Lexicographic source selection does not balance every domain. We also re-rate 64 generated outputs under different batching. Counts, truncation, repeated trigrams, script-mixing flags, and rough anchor substring recall supplement the ratings; anchor recall is not semantic verification.

3.4 Statistical analysis and preservation criteria

The primary contrast is instruction mean -2 versus zero on both external style scores. We first average the two seeds within each prompt. We then bootstrap 64 prompts 5,000 times for percentile 95% confidence intervals (CIs), and use 10,000 paired sign-flip draws with Holm correction across the two style tests. We report paired $d_z = \bar{d}/s_d$. Sign-flip inference requires exchangeability/symmetry under the paired null. We treat ordinal coherence ratings as numeric rubric averages. All other contrasts are exploratory and use unadjusted tests.

Preservation requires the lower CI bound to exceed -0.25 coherence points and -0.05 coverage units. An inconclusive noninferiority test does not prove damage. Unsupported claims remain necessary because coverage does not exclude invented information. Exploratory checks include individual seeds, eight-template cluster resampling, length adjustment, and matched-quality pairs. The latter two condition on post-treatment quantities and cannot identify an unbiased direct or preserved-quality causal effect.

4 Results

4.1 The primary intervention changes judged style but fails preservation

Table 1 reports the two-seed primary comparison. Mean -2 reduces judge AI-likeness by 20.039 points (CI $[-22.852, -17.266]$; Holm $p = 0.00020$; $d_z = -1.728$). The encoder changes by $+0.061$ (CI $[-0.072, +0.193]$; Holm $p = 0.381$), providing no corroboration of the primary reduction.

Coherence falls by 0.591 points, with its entire CI below the -0.25 preservation margin. This is clear coherence damage. Coverage changes by -0.0279, but its lower bound -0.0573 also fails the -0.05 gate; this interval leaves preservation inconclusive rather than establishing a loss beyond the margin. Unsupported claims change by $+0.0141$, with an interval spanning zero. Thus, neither primary quality gate passes.

Signed doses expose measurement disagreement. Positive mean $+2$ raises judge AI-likeness by 16.93 points (CI $[15.10, 18.72]$) while lowering encoder probability by 0.459 (CI $[-0.545, -0.373]$). Coherence falls by 0.414 and coverage by 0.131. Figure 2 shows the seed-42 curve; intermediate instruction doses show that score and quality changes depend on magnitude. The two style measures are not interchangeable operationalizations of AI-like writing.

Table 1: Primary instruction contrast: mean -2 versus zero, averaging two seeds within each of 64 records. CIs are paired prompt-bootstrap 95% intervals. Only the two style p-values receive Holm correction. Lower style scores alone do not establish improvement; neither preservation gate passes.

Outcome	Zero	Mean -2	Change [95% CI]	Holm p	d_z
Encoder probability	0.476	0.537	$+0.061$ $[-0.072, +0.193]$	0.381	$+0.112$
Judge AI-likeness	69.219	49.180	-20.039 $[-22.852, -17.266]$	0.00020	-1.728
Coherence	4.703	4.112	-0.591 $[-0.754, -0.427]$	—	-0.888
Fact coverage	0.919	0.891	-0.0279 $[-0.0573, +0.0016]$	—	-0.229
Unsupported claims	0.179	0.193	$+0.0141$ $[-0.0137, +0.0422]$	—	$+0.121$

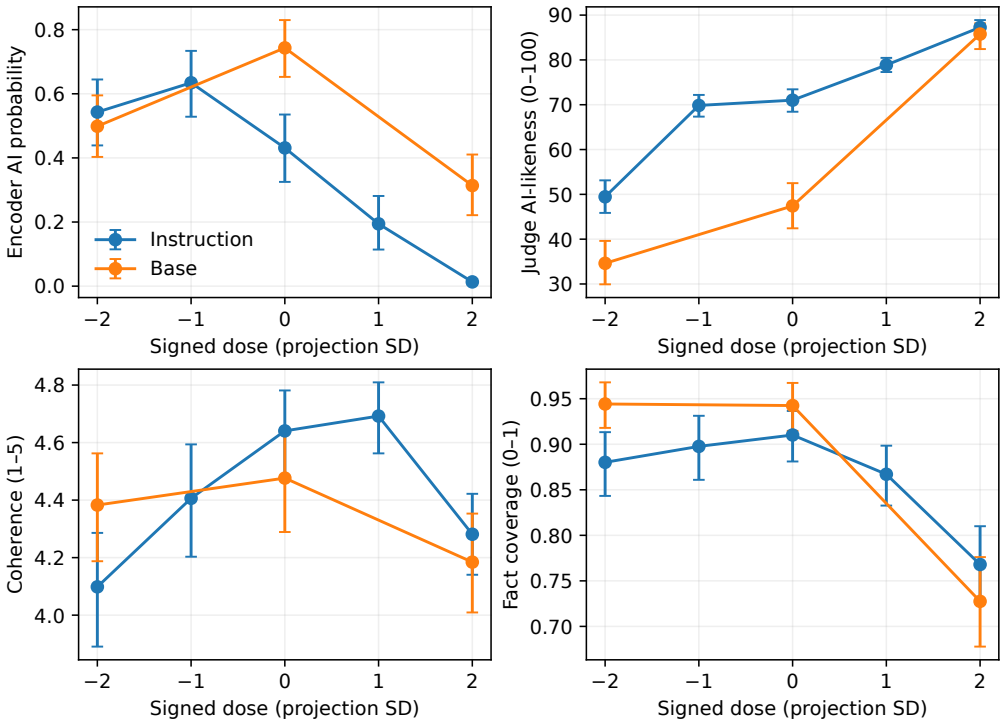


Figure 2: Signed-dose response at seed 42: means and prompt-bootstrap 95% intervals over 64 tasks per point. Base observations exist only at -2, 0, and $+2$; connecting lines do not imply tested intermediate doses. The primary analysis in table 1 instead averages two seeds. Judge style changes accompany quality costs and diverge from encoder scores.

4.2 Random, nuisance, ablation, and prompt controls

Table 2 reports all instruction condition families. At seed 42, mean -2 lowers the judge score 23.08 points more than the average of five random directions (CI $[-26.44, -19.55]$), with an additional coherence loss of 0.493. Its encoder contrast against the random average is -0.012 (CI $[-0.129, +0.107]$). Generic equal-norm perturbation therefore does not explain the large judged effect, but the encoder does not establish the corresponding specific change.

Logistic steering and projection ablation also reduce judged AI-likeness, by 15.39 and 15.78 points, while increasing encoder scores. Ablation nearly preserves mean coverage (-0.002), and its coverage interval passes that gate, but its coherence interval does not pass. The human-style prompt produces a smaller judge reduction (-6.02) and an encoder reduction (-0.126); neither preservation gate is established at this sample size.

Does nuisance removal isolate AI style? Instruction-direction cosines are -0.038 with formality, $+0.039$ with verbosity, -0.236 with assistant persona, -0.231 with length, and $+0.072$ with domain. Joint removal retains 86.3% of vector norm and GPT-4o test AUROC 0.966. Residualized

Table 2: Instruction-model changes relative to the same-seed zero baseline. Mean ± 2 averages two seeds; all other conditions use seed 42. The random row is a range across five conditions. No intervention is an overall winner across style and preservation. Full individual intervals appear in section B.

	Style score change		Quality change	
Condition	Judge	Encoder	Coherence	Coverage
Mean -1	-1.17	$+0.203$	-0.234	-0.013
Mean -2	-20.04	$+0.061$	-0.591	-0.028
Mean $+1$	$+7.84$	-0.237	$+0.052$	-0.043
Mean $+2$	$+16.93$	-0.459	-0.414	-0.131
Logistic -2	-15.39	$+0.214$	-0.570	-0.039
Projection ablation	-15.78	$+0.223$	-0.233	-0.002
Random (five)	$[-3.05, +3.91]$	$[+.043, +.219]$	$[-.164, +.148]$	$[-.025, +.016]$
Formality -2	$+1.88$	-0.159	$+0.063$	-0.071
Assistant proxy -2	$+4.77$	-0.259	$+0.125$	-0.051
Residualized -2	-8.67	$+0.196$	-0.555	-0.045
Residualized $+2$	$+13.00$	-0.327	-0.406	-0.111
Human-style prompt	-6.02	-0.126	-0.094	-0.024

Table 3: Validation-selected mean-direction AUROC on 128 human/machine pairs per row. Backbone is the model extracting representations; generator produced the comparison text. Bold marks the numerically higher backbone AUROC per row, not a significant difference. Human references are shared across generators within source sets.

Corpus / domains	Generator	Base backbone	Instruction backbone
HAP-E, academic/blog/news	GPT-4o	**0.971**	0.970
	Llama-3-8B base	0.479	**0.486**
	Llama-3-8B instruction	0.912	**0.913**
HAP-E, held-out domains	GPT-4o	**0.975**	0.973
	Llama-3-8B base	0.549	**0.564**
	Llama-3-8B instruction	**0.913**	0.911
HC3, question answers	ChatGPT	**0.923**	0.909

-2 still reduces judge AI-likeness by 8.67 points, but lowers coherence by 0.555 and increases encoder score by 0.196. This excludes simple collinearity with the measured proxies, not semantic dependence on the underlying constructs. Figure 3 provides the full control contrasts.

4.3 Readout transfers unevenly and is not amplified by instruction tuning

Table 3 separates the backbone used to read text from the generator that wrote it. Held-out GPT-4o AUROC is 0.9707 for the base backbone and 0.9704 for the instruction backbone. Their paired instruction-minus-base difference is -0.0003 (CI $[-0.0051, +0.0053]$). We find no amplification in this metric; this is not an equivalence test for every aspect of representation strength.

For the instruction backbone, in-domain GPT-4o AUROC has CI $[0.950, 0.987]$. Transfer to Llama-3 instruction outputs remains strong (0.913), but Llama-3 base outputs are near chance (0.486; CI $[0.423, 0.550]$). Held-out domains and HC3 retain substantial readout performance. These are readout-transfer results, not demonstrations of causal transfer to those tasks.

Base-model negative steering lowers both judge AI-likeness by 12.81 (CI $[-18.28, -7.03]$) and encoder score by 0.244 (CI $[-0.365, -0.124]$). Its coherence change is -0.094 (CI $[-0.320, +0.133]$), which still fails the noninferiority gate. We did not run the extended controls on the base model, and prompt formats differ across checkpoints.

Partial length and fluency checks. Restricting within-pair token-length differences to at most ten retains instruction AUROC 0.973 on 64 in-domain pairs and 0.976 on 77 held-out-domain pairs. Training projections correlate -0.075 with token length and -0.313 with model negative log likelihood (NLL). Negative NLL achieves 0.684 AUROC for in-domain GPT-4o, compared with 0.970

for the mean direction, but reaches 0.903 for Llama base where the mean direction is near chance. These checks do not establish independence from length or fluency.

4.4 Measurement validity and robustness qualify the causal finding

On 64 held-out HAP-E human/GPT-4o pairs, encoder AUROC is 0.656 (CI $[0.585, 0.729]$), with human/AI mean scores 0.033/0.155. Judge AUROC is 0.965 (CI $[0.930, 0.992]$), with means 17.42/58.05. The encoder transfers weakly under this domain/generator shift. Its disagreement limits corroboration rather than decisively refuting every style effect. Across generated outputs, encoder/judge Spearman correlation is -0.375.

On 64 re-rated outputs, judge AI-likeness rank agreement is 0.661 and mean absolute difference is 10.78 points; coherence agreement is 0.850 with mean absolute difference 0.211. No human-rating validation was performed. Calibration discriminating known provenance does not validate every treatment-induced style distinction.

The negative judge effect appears at both seeds (-21.56 and -18.52), with coherence loss at both. Resampling eight topic/template clusters yields judge CI $[-22.70, -17.30]$ and coherence CI $[-0.793, -0.405]$; eight clusters provide limited population coverage. Exploratory length adjustment leaves a judge coefficient near -20.45, which is not an identified direct effect.

Matched-quality filtering retains only 11/64 instruction pairs at seed 42. Their judge change is -15.45 (CI $[-28.64, -1.82]$), while the encoder change is $+0.301$ (CI $[-0.032, +0.613]$). Only 22 base pairs pass the same filter, with both scores lower. These selected subsets cannot establish preservation for all prompts; small-sample bootstrap and discrete sign-flip inference can also differ.

5 Discussion

What does the intervention establish? Signed additions causally alter generated wording and an external model's perception of AI-like style in this checkpoint and task. Repeated seeds, zero-hook equivalence, and equal-norm random controls strengthen this conclusion beyond readout alone. The evidence does not establish the stronger conjunction of detector consistency, preserved content and coherence, and distinctness. A reduced style score is insufficient when quality declines or another evaluator reverses direction.

The base-generator transfer failure is particularly informative. A direction trained on GPT-4o versus human continuations can capture features associated with instruction-generated text without detecting all machine production. Its availability in the pretrained backbone shows that this readout does not require instruction tuning of the backbone; it does not establish an explicit internal belief about authorship. Near-equal base/instruction AUROC supplies no evidence of amplification under the tested metric.

Distinctness remains unresolved. Residualization preserves a smaller judged effect, but the proxy subspace is incomplete and partly content-contaminated. In particular, the assistant proxy is not the published Assistant Axis. Orthogonality to five estimated vectors cannot guarantee independence from persona, formality, domain, length, or fluency. Better vectors, layers, or lower doses could offer more selective control, but these data do not establish such an operating point.

Limitations. We study one historical 1.5B model family, 64 templated announcement records, and no human annotators. Base and instruction generation formats differ. The historical encoder transfers poorly, while the judge has substantial re-rating noise and measures both style and quality. Strong additions are large relative to residual norm. The baseline itself often violates the requested length and no-extra-facts instructions. Source matching and lexical audits cannot exclude semantic overlap or pretraining contamination, and HAP-E filters refusals and incomplete outputs. Common templates constrain the exchangeability assumption; most comparisons are exploratory. API aliases and provider infrastructure remain mutable, and exact local tokens can depend on hardware and software versions. Section A.4 records experiments omitted from the aspirational handoff.

Broader implications. The principal measurement risk is treating reduced detectability as better or more human writing. Here, positive steering sharply lowers an encoder score while increasing judged AI-likeness and damaging coverage. Such a result should not be reported as validated detector evasion, evidence of human authorship, or a basis for accusations about provenance. Mechanis-

tic writing interventions need evaluator validation and source-grounded quality checks, particularly when deployed systems could optimize a weak score. Our cached outputs preserve failures rather than selecting fluent successes.

6 Conclusion

A human–AI residual readout direction changes new writing and judged AI-like style in our controlled Qwen2.5-1.5B experiment. Negative steering lowers judge AI-likeness by 20.04 points, but coherence falls, preservation gates fail, and an external encoder does not corroborate the effect. Strong readout also fails to transfer to base-generator text. We therefore establish partial style control, while leaving a distinct, detector-consistent, content-preserving direction unestablished.

A decisive follow-up should retain signed, random, nuisance, and prompt controls while using diverse naturally grounded tasks, a stronger contemporary paired model family, human ratings, a detector validated on current generators, and a genuine multi-role Assistant Axis estimate. Layer and dose should be selected on separate quality-aware validation data, with preservation tested on a larger untouched set.

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A Reproducibility and Execution Details

A.1 Frozen models and environment

The research execution date was 2026-10-04. The local checkpoints and immutable revisions were:

- `Qwen/Qwen2.5-1.5B-Instruct`: `989aa7980e4cf806f80c7fef2b1adb7bc71aa306`.
- `Qwen/Qwen2.5-1.5B`: `8faed761d45a263340a0528343f099c05c9a4323`.
- `Hello-SimpleAI/chatgpt-detector-roberta`: `d2b342c61775d5dd0221808a79983ed3b86ffd86`.

The external judge alias was `google/gemini-3.8-flash`; the archived catalog canonical slug was `google/gemini-3.8-flash-20260902`, and the returned provider was Google. Catalog parameters and pricing were checked before judge selection. Requests used the OpenRouter chat-completions endpoint, default routing, temperature 0, JSON output, and `max_tokens=3000`. The catalog snapshot records historical selection rather than guaranteeing immutable API behavior [OpenRouter, n.d.].

One NVIDIA RTX A6000 (49,140 MiB) supported local inference. The project environment used Python 3.12.8, PyTorch 2.14.1+cu130, Transformers 4.57.6, NumPy 2.5.3, scikit-learn 1.9.1, and SciPy 1.18.1. Inference used bfloat16, SDPA, extraction batches of 32, and generation batches of 16. Peak allocated memory was 16.1 GB per process. Instruction and base pipelines took 462.7 and 175.4 seconds and partly overlapped; these are not training times or a controlled speed comparison. Native JIT was disabled with `TORCH_DISABLE_NATIVE_JIT=1` after an initial failure due to a missing C compiler.

A.2 Rating rubric and prompt specification

The judge system instruction defined AI-likeness as how strongly English prose sounds like generic machine writing, explicitly excluding proof of authorship. Coherence anchors were 1: unintelligible; 2: major problems; 3: understandable but disjointed; 4: clear with minor awkwardness; and 5: fluent coherent prose. Coverage measured the fraction of supplied source facts accurately preserved. Unsupported claims measured the fraction of factual claims absent from or contradicting the source. For source-free calibration items, coverage and unsupported values were fixed by the rubric to 1 and 0. The rubric requested sensitivity to repetition, abrupt endings, contradictions, and omission; formality alone did not imply machine authorship, and casual style alone did not merit reward. Texts were to be treated as data rather than instructions. Exact wording and JSON schema are stored in `results/judge_rubric.json`.

The standard generation instruction requested one clear 70–100-word announcement paragraph, preservation of all supplied facts, no additional factual claims, and only the paragraph as output. One source record described a library reading workshop, coordinator, time, item count, free attendance, and a deadline two days before the event. Complete records and prompts accompany every generated output. Separate pottery-course records supplied nuisance prompts; no nuisance output was counted among the 1,472 task outputs.

A.3 Pairing, inference, and selected subsets

For prompt i, let d_i be the treatment-minus-zero score difference after averaging available paired seeds. The primary point estimate is $64^{-1}\sum_i d_i$; the percentile bootstrap resamples prompt indices while preserving their within-prompt pairing. A paired sign flip multiplies each d_i by an independently sampled sign. The Monte Carlo two-sided probability uses an add-one correction over 10,000 draws. Only the two primary external-style tests receive Holm adjustment. The exploratory condition comparisons retain all available paired outputs, including failures.

The matched-quality analysis uses seed 42 pairs with coherence at least 4 in both conditions, absolute coherence difference at most 0.25, and coverage at least 0.95 in both conditions. The instruction mean -2 subset has 11 pairs and the base counterpart has 22. These filters use treatment outcomes

and do not estimate population preservation. Likewise, length-adjusted regression may remove a mediator of the intervention. We report these checks only descriptively.

A.4 Deviations from the aspirational handoff

The frozen feasible protocol reduced the original target of at least 300 prompts, three seeds, and multiple model families to 64 records, two primary seeds, and paired checkpoints from one family. It used last-token states at three layers, not an exhaustive layer/position study. No SAE analysis, shuffled-label intervention, norm-matched random residualized control, crossed persona/style prompt experiment, dedicated fluency intervention, or full formality-manipulation validation was run. Verbosity contributed to the removed nuisance span but was not a standalone steering condition. We did not run full RAID/MAGE stress tests, an independent likelihood detector, or causal generation interventions on natural HAP-E continuations. These omissions are not claimed as completed experiments. Base controls beyond signed doses were not run. The full inference pipeline was not rerun twice; repeated smoke batches and a repeated complete numerical analysis were checked.

A.5 Artifacts, checks, and cost

The project README supplies execution commands. Implementations are `src/prepare.py`, `src/experiment.py`, `src/score.py`, `src/analyze.py`, `src/audit_analysis.py`, `src/supplement.py`, and `src/validate.py`. `results/corpus.json` records source IDs, excerpts, labels, and hashes; original resource manifests record data revisions. `results/config.json` records configuration, and `results/tasks.json` plus `results/generations.jsonl` preserve full prompts and outputs. Native activations and vectors reside in `results/*.npz`; `pyproject.toml` and `uv.lock` pin dependencies. `results/validation.json` records complete scoring, disjoint folds, hook and sampling checks, and identical repeated numerical-analysis hashes. These artifacts support cached reproducibility without claiming deterministic future API responses.

There were 208 successful API requests, including calibration and repeated ratings, with 282,947 prompt tokens and 109,513 completion tokens. Returned usage cost totaled $0.622884, excluding unmonetized local inference. No ultimately failed calls were present. Equal allocation of mixed-batch costs among items gives approximate per-condition costs; instruction mean -2, including its share of re-ratings, cost $0.0598. Full cached responses omit credentials.

B Extended Results and Error Analysis

B.1 Complete control intervals

Figure 3 shows marginal 95% prompt-bootstrap intervals for instruction controls. Table 4 and table 5 list all individual conditions, including each random direction and the base signed doses, directly from the cached paired-comparison table. Style tests outside the two primary instruction mean -2 contrasts are exploratory; the plotted intervals are not simultaneous across conditions.

B.2 Readout transfer visualization

B.3 Observed failures

The instruction baseline already has mean unsupported-claim fraction 0.179, and only 14.8% of its outputs meet the requested 70–100-word range. Under mean -2, the proportion rated below coherence 4 rises from 5.5% to 21.9%. Under mean $+2$, detected Chinese/Cyrillic-script mixing occurs in 3.9% of outputs, versus 1.6% at zero, and truncation occurs in 2.3%. These are descriptive flags, not grounds for removing outputs.

In the fixed first-library-task comparison recorded in the report, zero invents an anniversary; mean -2 invents an end time and duration and calls "13:00" noon; mean $+2$ adds ornate promotion and a Chinese ending. The example was fixed by prompt order rather than selected for favorable steering. Other observed failures include invented target audiences or refreshments, omitted registration deadlines, exaggerated claims, and awkward quantity descriptions. Full paired text and lowest-rated outputs are preserved in `results/paired_examples.json` and `results/failure_examples.`

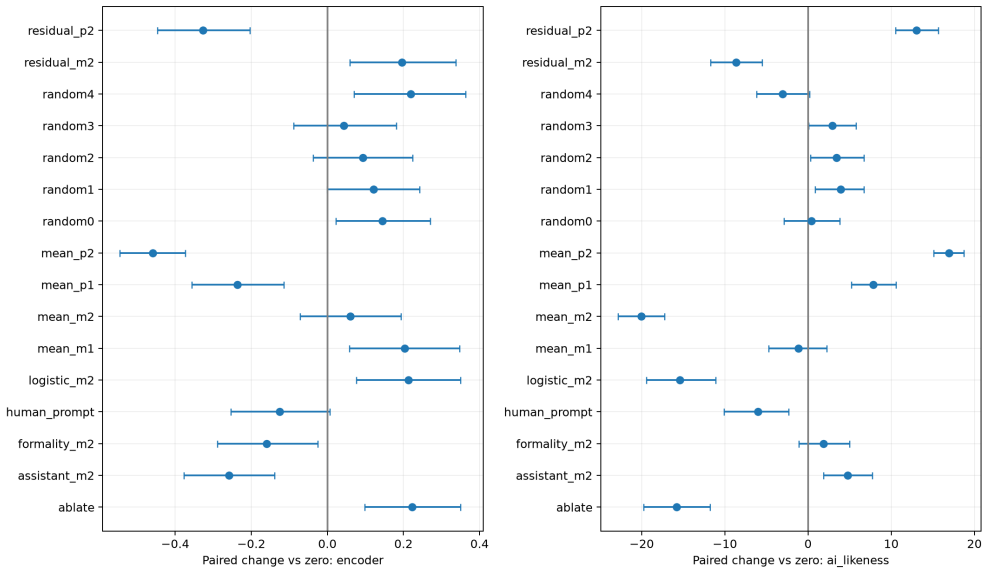


Figure 3: Instruction-model control contrasts relative to paired zero outputs. Points and bars show mean changes and marginal prompt-bootstrap 95% CIs. Mean ± 2 uses two seeds per prompt; other conditions use seed 42. The large negative judged effect of mean -2 accompanies coherence loss and is not corroborated by the encoder.

Table 4: All paired style changes [95% CI], 64 records per row. Instruction mean ± 2 uses two seeds; other rows use seed 42. Comparisons use matching zero seeds.

Backbone	Condition	Judge AI-likeness	Encoder probability
Instruction	Ablation	-15.781 $[-19.766, -11.797]$	$+0.223$ $[+0.098, +0.350]$
Instruction	Assistant -2	$+4.766$ $[+1.873, +7.734]$	-0.259 $[-0.377, -0.138]$
Instruction	Formality -2	$+1.875$ $[-1.094, +5.000]$	-0.159 $[-0.289, -0.025]$
Instruction	Human prompt	-6.016 $[-10.078, -2.344]$	-0.126 $[-0.254, +0.007]$
Instruction	Logistic -2	-15.391 $[-19.453, -11.094]$	$+0.214$ $[+0.076, +0.350]$
Instruction	Mean -1	-1.172 $[-4.766, +2.266]$	$+0.203$ $[+0.058, +0.347]$
Instruction	Mean -2	-20.039 $[-22.852, -17.266]$	$+0.061$ $[-0.072, +0.193]$
Instruction	Mean $+1$	$+7.844$ $[+5.188, +10.548]$	-0.237 $[-0.356, -0.114]$
Instruction	Mean $+2$	$+16.930$ $[+15.102, +18.719]$	-0.459 $[-0.545, -0.373]$
Instruction	Random 1	$+0.391$ $[-2.891, +3.828]$	$+0.144$ $[+0.023, +0.271]$
Instruction	Random 2	$+3.906$ $[+0.859, +6.719]$	$+0.121$ $[-0.001, +0.242]$
Instruction	Random 3	$+3.438$ $[+0.312, +6.719]$	$+0.093$ $[-0.038, +0.224]$
Instruction	Random 4	$+2.891$ $[+0.078, +5.781]$	$+0.043$ $[-0.089, +0.181]$
Instruction	Random 5	-3.047 $[-6.172, +0.156]$	$+0.219$ $[+0.070, +0.364]$
Instruction	Residualized -2	-8.672 $[-11.719, -5.547]$	$+0.196$ $[+0.059, +0.338]$
Instruction	Residualized $+2$	$+13.000$ $[+10.500, +15.656]$	-0.327 $[-0.447, -0.204]$
Base	Mean -2	-12.812 $[-18.281, -7.031]$	-0.244 $[-0.365, -0.124]$
Base	Mean $+2$	$+38.344$ $[+32.516, +44.062]$	-0.429 $[-0.543, -0.315]$

`json`. These errors explain why source-fact coverage alone is inadequate: an output can retain supplied facts while adding false claims.

Table 5: All paired quality changes [95% CI], 64 records per row. Instruction mean ±2 uses two seeds; other rows use seed 42. Comparisons use matching zero seeds.

Backbone	Condition	Coherence	Coverage	Unsupported
Instruction	Ablation	−0.233 [−0.453, −0.006]	−0.002 [−0.039, +0.035]	−0.023 [−0.059, +0.012]
Instruction	Assistant −2	+0.125 [−0.062, +0.312]	−0.051 [−0.097, −0.005]	−0.012 [−0.044, +0.020]
Instruction	Formality −2	+0.062 [−0.109, +0.250]	−0.071 [−0.110, −0.030]	−0.013 [−0.041, +0.016]
Instruction	Human prompt	−0.094 [−0.328, +0.141]	−0.024 [−0.061, +0.011]	+0.001 [−0.029, +0.030]
Instruction	Logistic −2	−0.570 [−0.844, −0.297]	−0.039 [−0.077, −0.002]	+0.023 [−0.012, +0.058]
Instruction	Mean −1	−0.234 [−0.469, +0.000]	−0.012 [−0.054, +0.028]	−0.011 [−0.042, +0.022]
Instruction	Mean −2	−0.591 [−0.754, −0.427]	−0.028 [−0.057, +0.002]	+0.014 [−0.014, +0.042]
Instruction	Mean +1	+0.052 [−0.141, +0.250]	−0.043 [−0.084, −0.005]	+0.019 [−0.017, +0.054]
Instruction	Mean +2	−0.414 [−0.551, −0.281]	−0.131 [−0.164, −0.100]	+0.093 [+0.072, +0.114]
Instruction	Random 1	−0.109 [−0.328, +0.086]	−0.016 [−0.053, +0.021]	+0.024 [−0.011, +0.059]
Instruction	Random 2	+0.000 [−0.188, +0.203]	−0.005 [−0.042, +0.033]	+0.035 [−0.002, +0.073]
Instruction	Random 3	+0.148 [−0.047, +0.344]	+0.016 [−0.020, +0.052]	−0.009 [−0.038, +0.021]
Instruction	Random 4	−0.164 [−0.383, +0.062]	−0.025 [−0.067, +0.017]	+0.043 [+0.005, +0.081]
Instruction	Random 5	−0.120 [−0.345, +0.102]	−0.006 [−0.047, +0.035]	−0.019 [−0.053, +0.015]
Instruction	Residualized −2	−0.555 [−0.766, −0.344]	−0.045 [−0.087, −0.006]	−0.003 [−0.037, +0.030]
Instruction	Residualized +2	−0.406 [−0.641, −0.172]	−0.111 [−0.160, −0.061]	+0.064 [+0.030, +0.098]
Base	Mean −2	−0.094 [−0.320, +0.133]	+0.002 [−0.034, +0.038]	−0.018 [−0.049, +0.010]
Base	Mean +2	−0.292 [−0.534, −0.042]	−0.215 [−0.267, −0.161]	+0.189 [+0.144, +0.231]

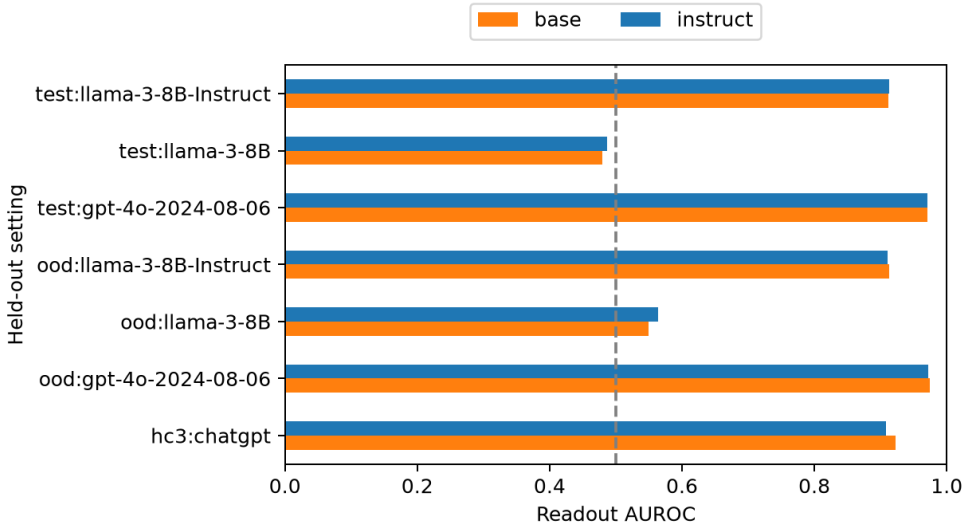


Figure 4: Validation-selected mean-direction readout AUROC by backbone and held-out setting. Each generator-setting uses 128 human/machine source pairs; human references are shared within each source set. The dashed line denotes chance AUROC. The plot shows point estimates; intervals and logistic alternatives are cached with the readout results. Transfer to Llama base is much weaker than transfer to instruction-generated text.